

Statistical Business Analysis Project

1. Project Overview and Objectives

Project Overview

This project performs a statistical analysis of business and customer data to uncover revenue trends, customer behavior, and contract/tenure patterns. It integrates descriptive statistics, hypothesis testing, correlation, confidence intervals, and regression modeling to generate actionable business insights for data-driven decision-making.

Objectives

- To analyze revenue, churn, and customer metrics using Python and Pandas
- To perform hypothesis tests to identify significant differences between groups
- To explore correlations and relationships among key business variables
- To calculate confidence intervals and regression models for predictive insights
- To provide clear, actionable business recommendations based on data

2. Setup and Installation Instructions

Prerequisites

- Python 3.10+
- Jupyter Notebook

Installation Steps

1. Download or clone the project repository from GitHub.
2. Install required libraries:

```
pip install -r requirements.txt
```
3. Place dataset inside /data folder.
4. Launch the notebook:

Python notebook

5. Open notebook/statistical_analysis.ipynb in Jupyter Notebook for analysis.
6. Check reports/hypothesis_tests_results.txt for saved hypothesis test outcomes.

3. Code Structure Explanation

```
Statistical_business_analysi_week7/  
|  
├── data/  
|   ├── customer_churn.csv  
|   ├── sales_data.csv  
|   └── business_data.csv  
├── notebook/  
|   └── statistical_analysis.ipynb  
├── reports/  
|   ├── statistical_report.pdf  
|   └── hypothesis_tests_results.txt  
├── create_business_data.py  
├── requirements.txt  
└── README.md
```

File Descriptions

- data/customer_churn.csv
Customer subscription and churn dataset.
- data/sales_data.csv
Sales transactions dataset with products, quantities, prices, and regions.
- data/business_data.csv
Cleaned and merged dataset for analysis.
- notebook/statistical_analysis.ipynb
Jupyter Notebook performing step-by-step statistical analysis and insights.

- reports/statistical_report.pdf
Professional report summarizing analysis results and business insights.
- reports/hypothesis_tests_results.txt
Saved output of all hypothesis test results.
- requirements.txt
Specifies all Python dependencies needed to run the project.
- README.md
Project documentation including overview, setup instructions, and usage guide.

4. Screenshots of Working Application

The included screenshots demonstrate:

- Total revenue distribution shows a right-skewed pattern, highlighting a few high-value customers.
- Descriptive statistics confirm data consistency and reveal central tendencies.
- OLS regression indicates that MonthlyCharges, Tenure, and SeniorCitizen are not statistically significant predictors for revenue ($R^2 = 0.006$), suggesting other factors influence total revenue.

OLS Regression Results			
Dep. Variable:	Total_Revenue	R-squared:	0.006
Model:	OLS	Adj. R-squared:	0.000
Method:	Least Squares	F-statistic:	1.013
Date:	Fri, 27 Feb 2026	Prob (F-statistic):	0.387
Time:	19:54:34	Log-Likelihood:	-6261.0
No. Observations:	500	AIC:	1.253e+04
Df Residuals:	496	BIC:	1.255e+04
Df Model:	3		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.333e+04	9975.504	1.336	0.182	-6274.049	3.29e+04
MonthlyCharges	7.2657	58.038	0.125	0.900	-106.765	121.296
Tenure	192.9534	144.767	1.333	0.183	-91.479	477.386
SeniorCitizen	7088.4872	6000.856	1.181	0.238	-4701.745	1.89e+04

Omnibus:	348.055	Durbin-Watson:	0.907
Prob(Omnibus):	0.000	Jarque-Bera (JB):	2895.957
Skew:	3.156	Prob(JB):	0.00
Kurtosis:	12.958	Cond. No.	449.

Fig. 4.1. Regression Analysis

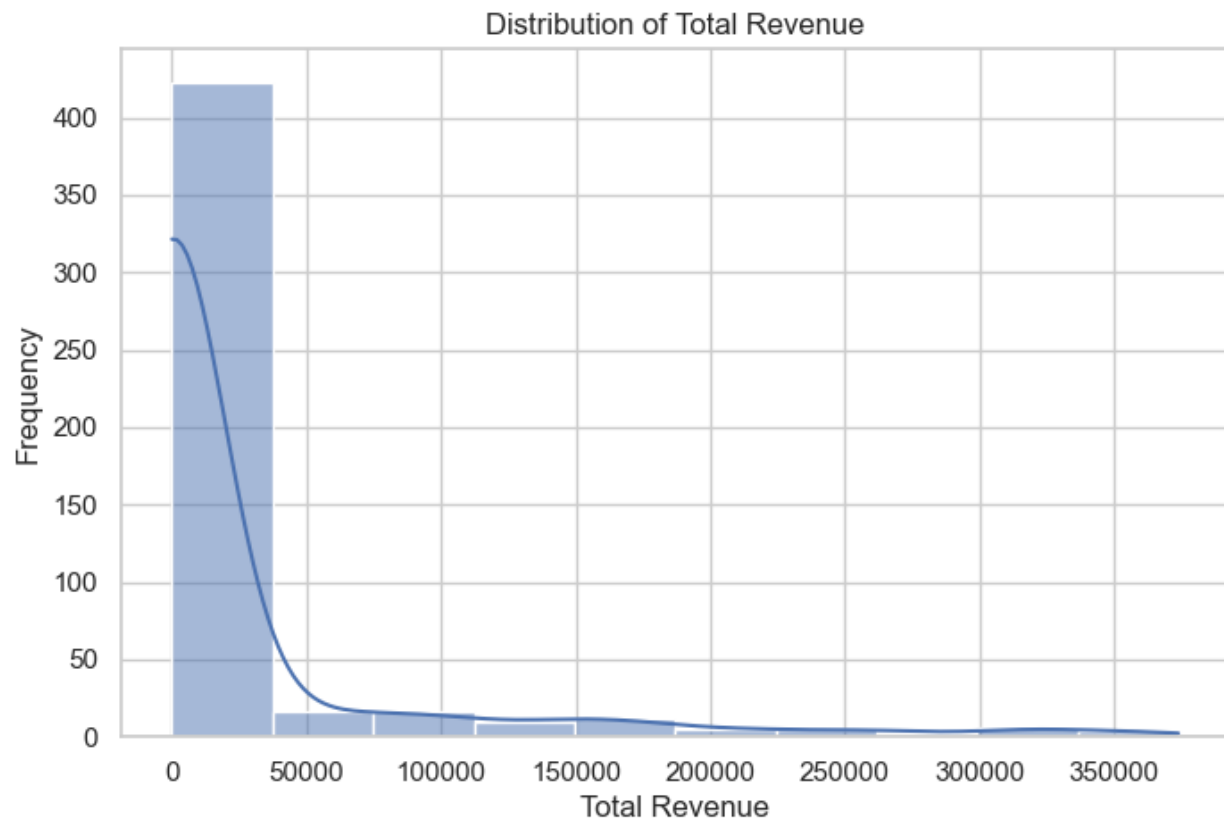


Fig. 4.2. Distribution Analysis

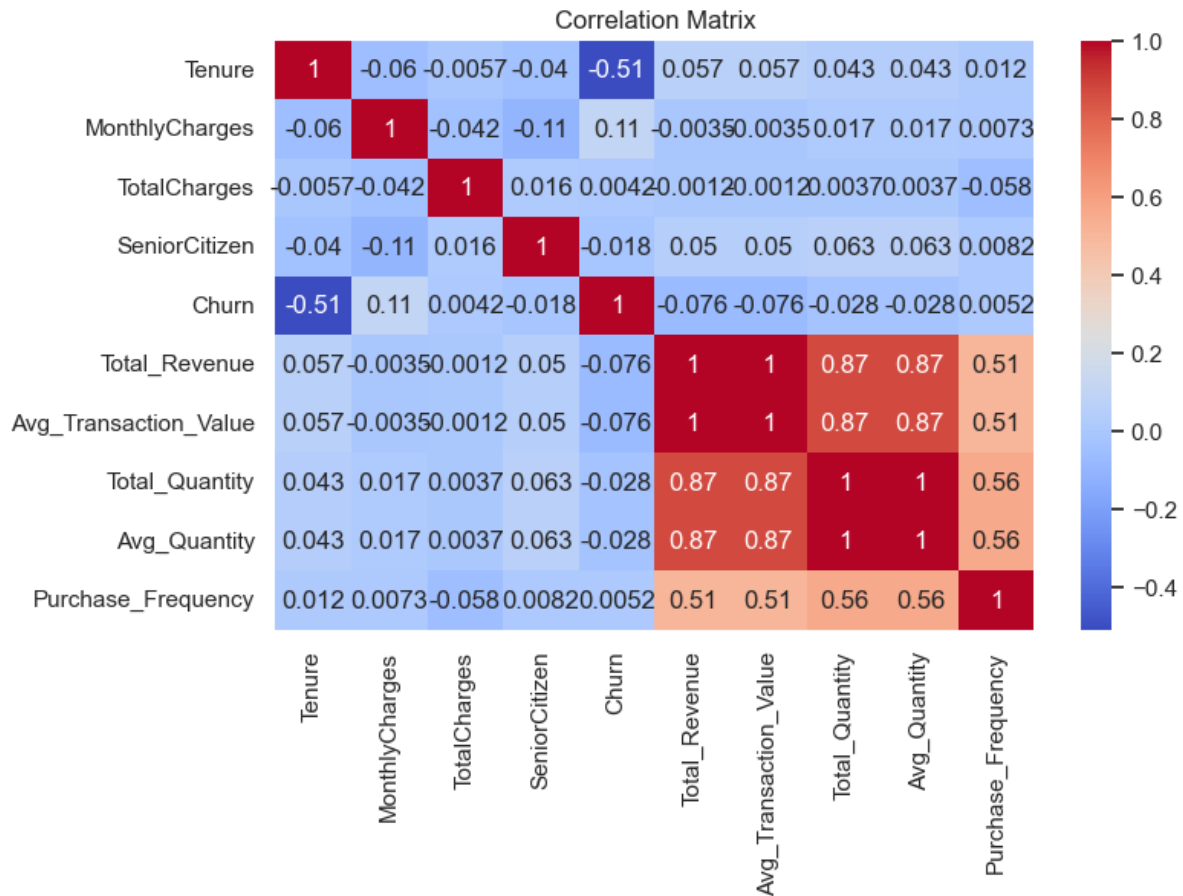


Fig. 4.3. Correlation Analysis

	Tenure	MonthlyCharges	TotalCharges	SeniorCitizen	Churn	Total_Revenue	Avg_Transaction_Value	Total_Quantity	Avg_Quantity	Purchase_Frequency
count	500.000000	500.000000	500.000000	500.000000	500.000000	500.000000	500.000000	500.000000	500.000000	500.000000
mean	36.532000	113.636000	4237.882000	0.498000	0.106000	24730.096000	24730.096000	0.956000	0.956000	0.090000
std	20.667057	51.799903	2260.619837	0.500497	0.308146	66645.163839	66645.163839	2.234289	2.234289	0.286468
min	1.000000	20.000000	159.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	19.000000	67.000000	2237.250000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
50%	37.000000	115.000000	4182.500000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
75%	54.000000	158.000000	6266.750000	1.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
max	71.000000	199.000000	7992.000000	1.000000	1.000000	373932.000000	373932.000000	9.000000	9.000000	1.000000

Fig. 4.4. Descriptive Statistics

These results validate that the statistical analysis was executed correctly and provide actionable insights for business decision-making.

5. Technical Requirements Fulfillment Explanation

Category	Technical Requirement	How It Was Implemented
Algorithm	Data loading and preprocessing	Sequential pipeline loads business_data.csv, handles missing values, parses dates, and standardizes datatypes using Pandas.
Algorithm	Data cleaning logic	Missing numeric values imputed with median, duplicates removed, and invalid entries coerced to NaN.
Algorithm	KPI and metric calculations	Aggregations (sum, mean, count, groupby) computed total revenue, avg. order value, total quantity, and purchase frequency.
Algorithm	Trend identification	Daily, monthly, and contract-based trends analyzed using groupby and visualized via histograms and density plots.
Algorithm	Customer segmentation	Customer-level metrics (total revenue, avg transaction) computed and visualized to identify high-value segments.
Algorithm	Product performance analysis	Product-wise revenue, quantity, and distribution analyzed using bar plots, box plots, and violin plots.

Category	Technical Requirement	How It Was Implemented
Data Structure	Tabular data handling	Pandas DataFrames efficiently stored and manipulated structured sales and churn data.
Data Structure	Metric storage	Aggregated metrics stored in DataFrames and Series for easy access and further analysis.
Architecture	Modular design	Code organized into notebook, preprocessing script, and separate visualization modules for reusability.
Architecture	Configuration management	Color palettes, plot styles, and themes centralized for consistent styling across visualizations.
Output	Interactive dashboard	Streamlit dashboard combines charts and metrics for interactive exploration of revenue, product, and customer trends.
Output	Static statistical plots	Seaborn plots (histogram, heatmap, box, violin) generated and saved for reporting and documentation.
Output	Business insights	Key metrics, correlations, hypothesis tests, and regression results summarized with actionable recommendations for stakeholders.

6. Learning Outcomes

Through this project, the following learning outcomes were achieved:

- Gained hands-on experience in performing end-to-end statistical analysis using Python and Pandas.
- Learned to clean, preprocess, and merge real-world business and customer datasets.
- Developed skills in calculating key metrics such as total revenue, average revenue, and customer churn statistics.
- Improved proficiency in data visualization using Seaborn and Matplotlib, creating correlation heatmaps, distribution plots, and regression insights.
- Strengthened understanding of hypothesis testing, confidence intervals, regression modeling, and generating actionable business recommendations.

7. Conclusion

This project demonstrated how statistical analysis can transform business data into actionable insights. By applying descriptive statistics, hypothesis testing, correlation analysis, confidence intervals, and regression modeling, we identified key drivers of revenue and churn, enabling data-driven recommendations to improve customer retention and business performance.